Late Delivery Risk Prediction

Executive Summary: This project predicts whether an order is likely to be delivered late using machine learning techniques.

Problem Statement

The objective is to identify orders at risk of late delivery before shipment using available decision-time information.

Dataset

DataCo Supply Chain Dataset containing 180,519 records and 46 features after preprocessing.

Models Used

• Logistic Regression
• Decision Tree
• Random Forest

Results

Random Forest achieved the highest accuracy on the prepared dataset. The exceptionally high accuracy suggests potential data leakage and should be investigated before deploying the model in production.

Deployment

A Streamlit application was developed to demonstrate model loading and prediction.

Conclusion

This project demonstrates an end-to-end machine learning workflow for predicting late delivery risk, including preprocessing, model training, evaluation, and deployment.